LDPC CODEC IP

LDPC Codec IP is a configurable forward-error-correction core designed for high-speed communication systems. It supports multiple code rates, a 672-bit block size, and iterative decoding with early termination to reduce latency. The fully synchronous, FPGA-compatible design provides high throughput (300+ Mbps) using efficient decoding algorithms and includes flexible interfaces, buffering, and adjustable input settings for reliable data transmission.

Introduction

The LDPC Codec IP is a high-performance error-correction core developed to enhance data integrity in digital communication systems where transmitted information may be affected by noise or interference. It uses an iterative decoding technique that allows the system to detect and correct errors efficiently, resulting in improved reliability and signal quality. The core is designed with a flexible architecture that supports configurable parameters such as code rate selection, decoding iterations, and input data formats, enabling it to meet diverse application requirements.

Built with a fully synchronous and FPGA-friendly design, the IP ensures smooth data handling through simple control interfaces and internal buffering for continuous data flow. Its optimized decoding approach helps achieve high throughput while reducing processing latency through features like early termination. Due to its adaptability, efficiency, and strong error-correction capability, the LDPC Codec IP is well suited for modern high-speed communication and data transmission environments.

Architecture

The architecture of the LDPC Codec IP is based on a fully synchronous, single-clock design that enables efficient and reliable decoding operations. It primarily consists of input buffering, parameter control logic, an iterative decoding engine, and output buffering. The decoding engine implements an optimized iterative algorithm that processes incoming soft-decision data in multiple cycles to progressively correct errors and achieve accurate output results. Internal control logic manages configuration parameters such as code rate, block length, and iteration count, ensuring flexible operation without interrupting data flow.

To support continuous high-speed processing, the architecture includes handshaking signals and dual-block buffering that allow new data to be accepted while previous blocks are still being decoded. This streaming-oriented structure helps maintain high throughput and minimizes data loss. Additionally, features like early termination logic reduce unnecessary iterations when parity conditions are met, improving overall efficiency and lowering decoding latency.

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Features

- Fully synchronous design using a single clock.
- Code Rate support for 1/2 and 3/4.
- Block length of 672.
- Number of Iterations is set to 4.
- Support for user-defined iteration is available.
- Configurable soft input bit width.
- Provides clock enable support at input and output.
- Early termination
- Easy-to-use interface with handshaking signals.
- Supports all FPGA devices.
- Maximum synthesizable frequency is 100MHz to 200 MHZ depending on target FPGA.
- More than 300 Mbps throughput at a block size of 672 @ 250MHz clk.
- Decoding algorithm: Offset Min-Sum
- Message Scheduling algorithm: Layered Belief Propagation (LBP)
- Enhancement Available (On Demand)
- Support for different Code-Rates/Block-Sizes.
- Support for different communication standards e.g. 802.11n, 802.11ac, 802.16e, etc.
- System-Generator Model and Test-bench availability (On demand)

Specification

GENERAL CONFIGURATION	
Parameter	**Specification**
Design Type	• Fully synchronous design using a single clock
Supported Code Rates	• 1/2, 3/4
Block Length	• 672 bits
Iterations	• Default = 4 (User-defined supported)
Input Bit Width	• Configurable soft input bit width
Clock Enable	• Supported at input and output
Early Termination	• Supported
Interface	• Handshaking based interface
Device Compatibility	• Supports all FPGA devices

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PERFORMANCE	
Parameter	**Specification**
Maximum Frequency	• 100 MHz – 200 MHz (depending on FPGA)
Maximum Throughput	• >300 Mbps @ 250 MHz
Throughput (@250 MHz)	• 328 Mbps
ALGORITHMS USED	
Parameter	**Specification**
Decoding Algorithm	• Offset Min-Sum
Scheduling Algorithm	• Layered Belief Propagation (LBP)
RESOURCE UTILIZATION (@250 MHZ)	
Resource	**Usage**
Max Block Size	• 672
LUTs	• 11K
Flip-Flops	• 8K
Block RAMs	• 12
ENHANCEMENTS (ON DEMAND)	
Feature	**Details**
Code Flexibility	• Support for different code rates and block sizes
Standards Support	• 802.11n, 802.11ac, 802.16e, etc.
Additional Models	• System Generator Model and Testbench availability

DELIVERABLES	
Item	**Availability**
Product Specification	• Provided
Design Files	• Provided
RTL-VHDL	• Optional
Reference Design	• Provided
System Generator Model	• Optional
MATLAB Bit-Accurate Model	• Optional
TOOLS & SUPPORT	
Category	**Details**
Simulation Tool	• ModelSim SE
Synthesis Tools	• Xilinx ISE / Altera Quartus
Support Period	• Three Month Support Provided

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